INT 422
Pattern Recognition
2024

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Lecture 6: Agenda

- **What is BNN & ANN**
- **ANN Component**
- **Architecture of ANN**
- **Neural Network Models**
- **Feedforward vs. Backpropagation**
- **Activation function Types**

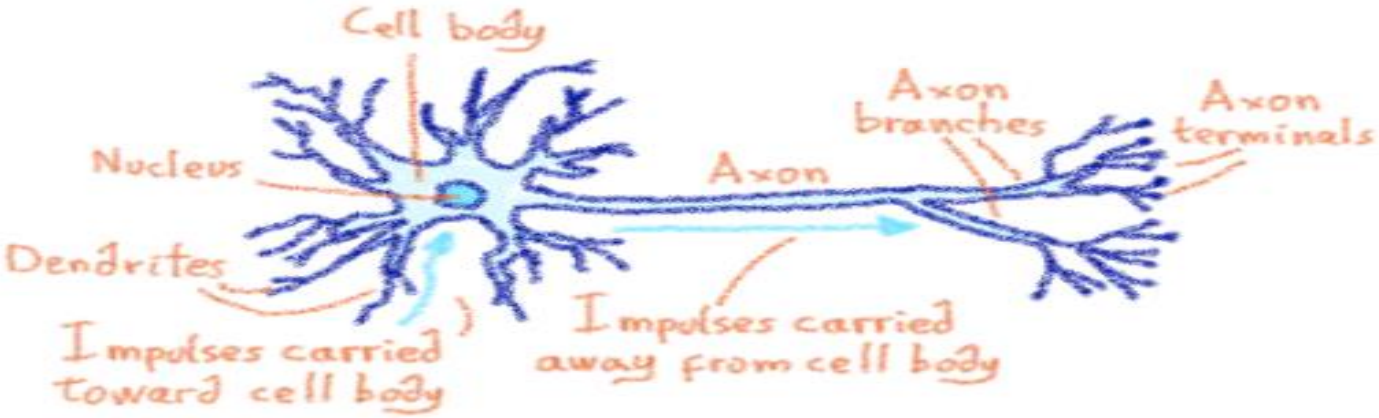
Introduction

- https://www.youtube.com/watch?v=rEDzUT3ymw4
- https://www.youtube.com/watch?v=2hMuYi2Wudw

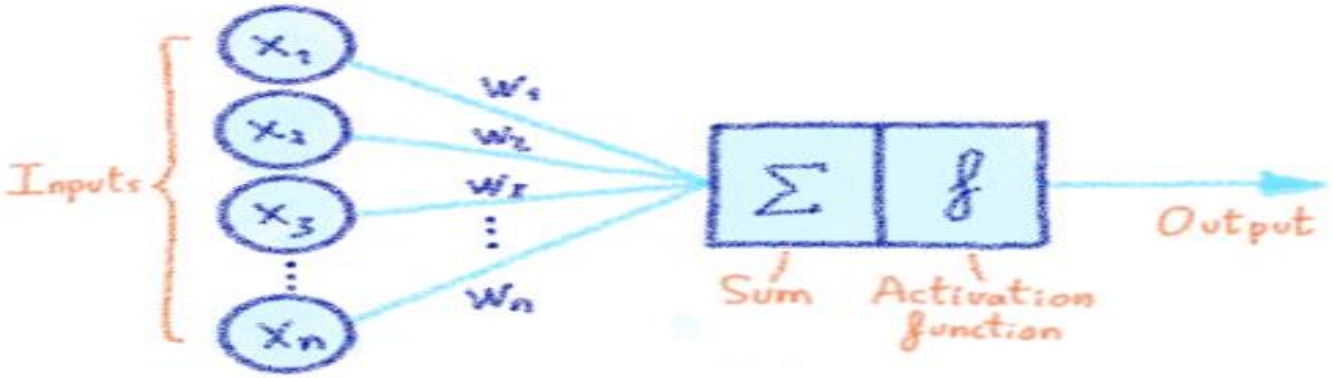
Introduction

- What is an (artificial) neural network
- A large set of **nodes** (units, neurons, processing elements)
  - Each node has input and output
  - Each node performs a **simple** computation by its node function
- **Weighted connections** between nodes
  - Connectivity gives the structure/architecture of the net
  - Connections/links have directions
  - What can be computed by a NN is primarily determined by the connections and their weights.

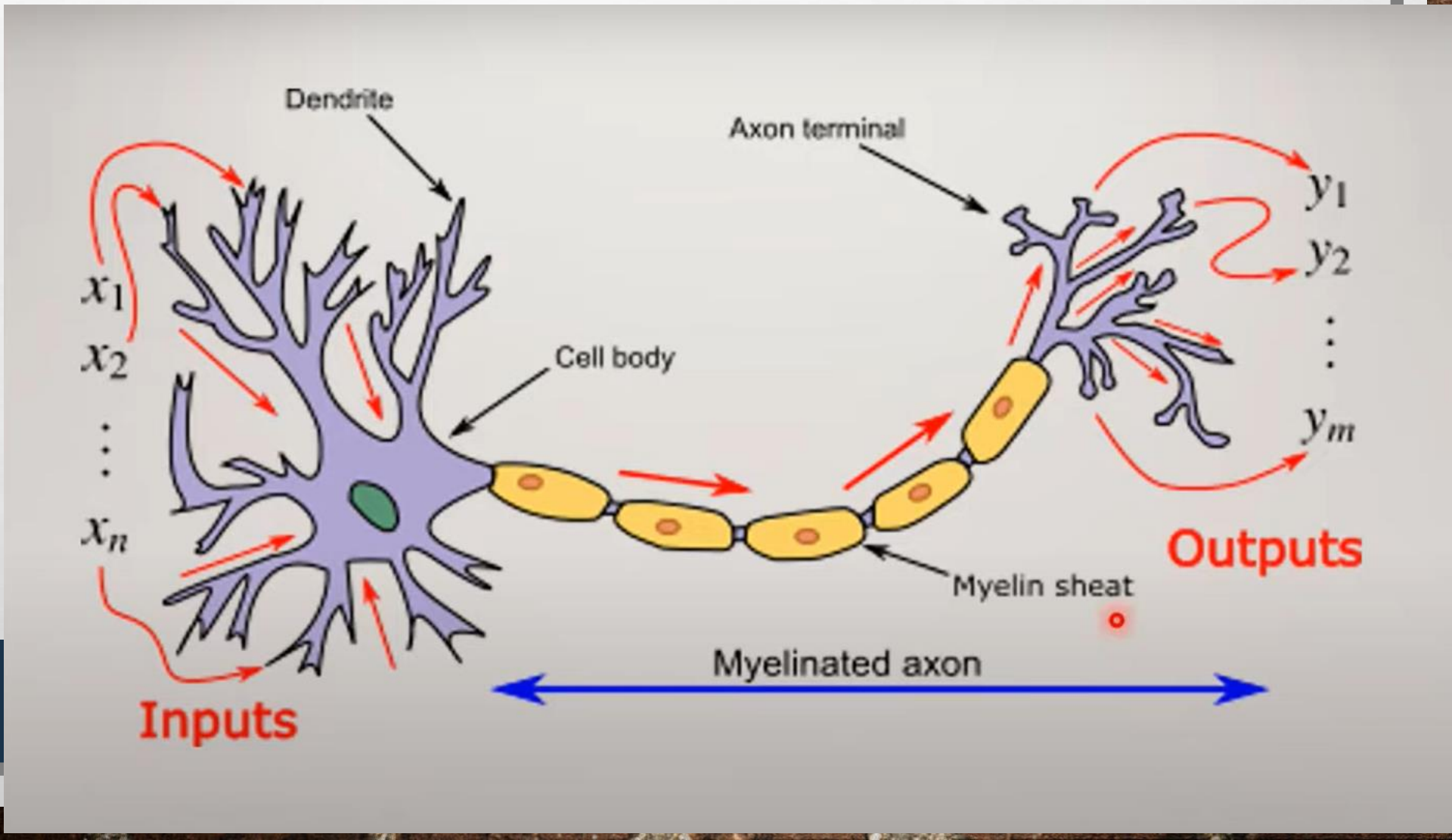
Biological Neuron



Artificial Neuron

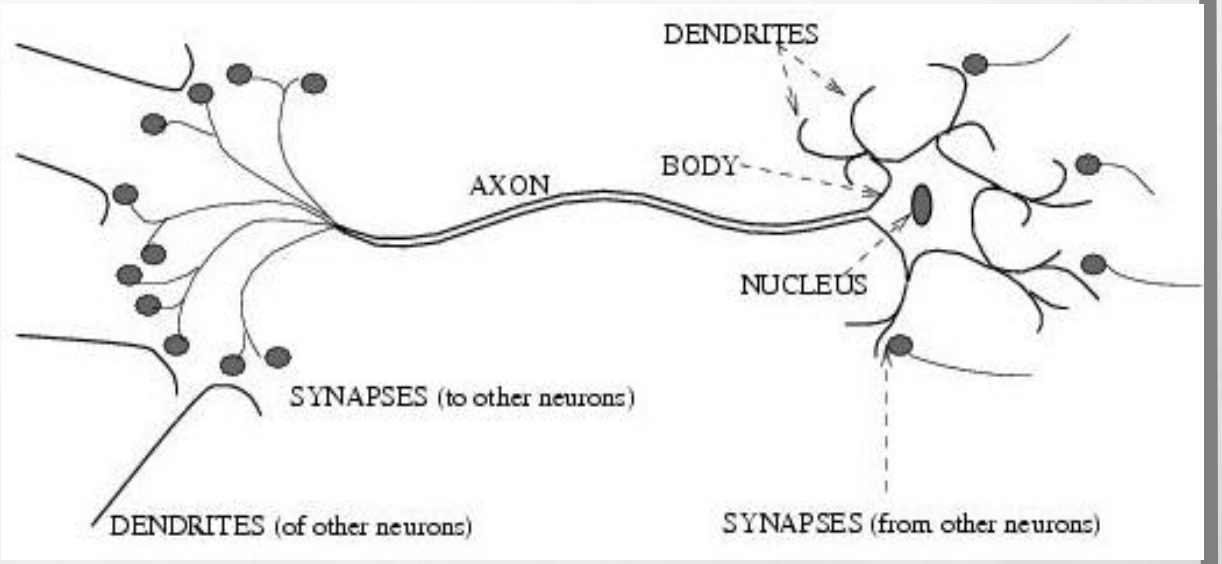


BNN & ANN


Dendrite
Axon terminal
x_1
x_2
x_n
Cell body
y_1
y_2
y_m
Outputs
Myelin sheat
Myelinated axon
Inputs

Biological Neural Activity

- Each neuron has a *body*, an *axon*, and many *dendrites*
- Can be in one of the two states: *firing* and *rest*.



- Neuron fires if the total incoming stimulus exceeds the threshold
- *Synapse*: thin gap between axon of one neuron and dendrite of another.
  - Signal exchange
  - Synaptic strength/efficiency

Introduction

ANN

- **Nodes**
  - input
  - output
  - node function
- **Connections**
  - connection strength

Bio NN

- **Cell body**
  - signals from other neurons
  - firing frequency
  - firing mechanism
- **Synapses**
  - synaptic strength

- Highly parallel, simple local computation (*at neuron level*) achieves global results as emerging property of the interaction (*at network level*)
- Pattern directed (meaning of individual nodes only in the context of a pattern)
- Fault-tolerant/graceful degrading
- Learning/adaptation plays important role.

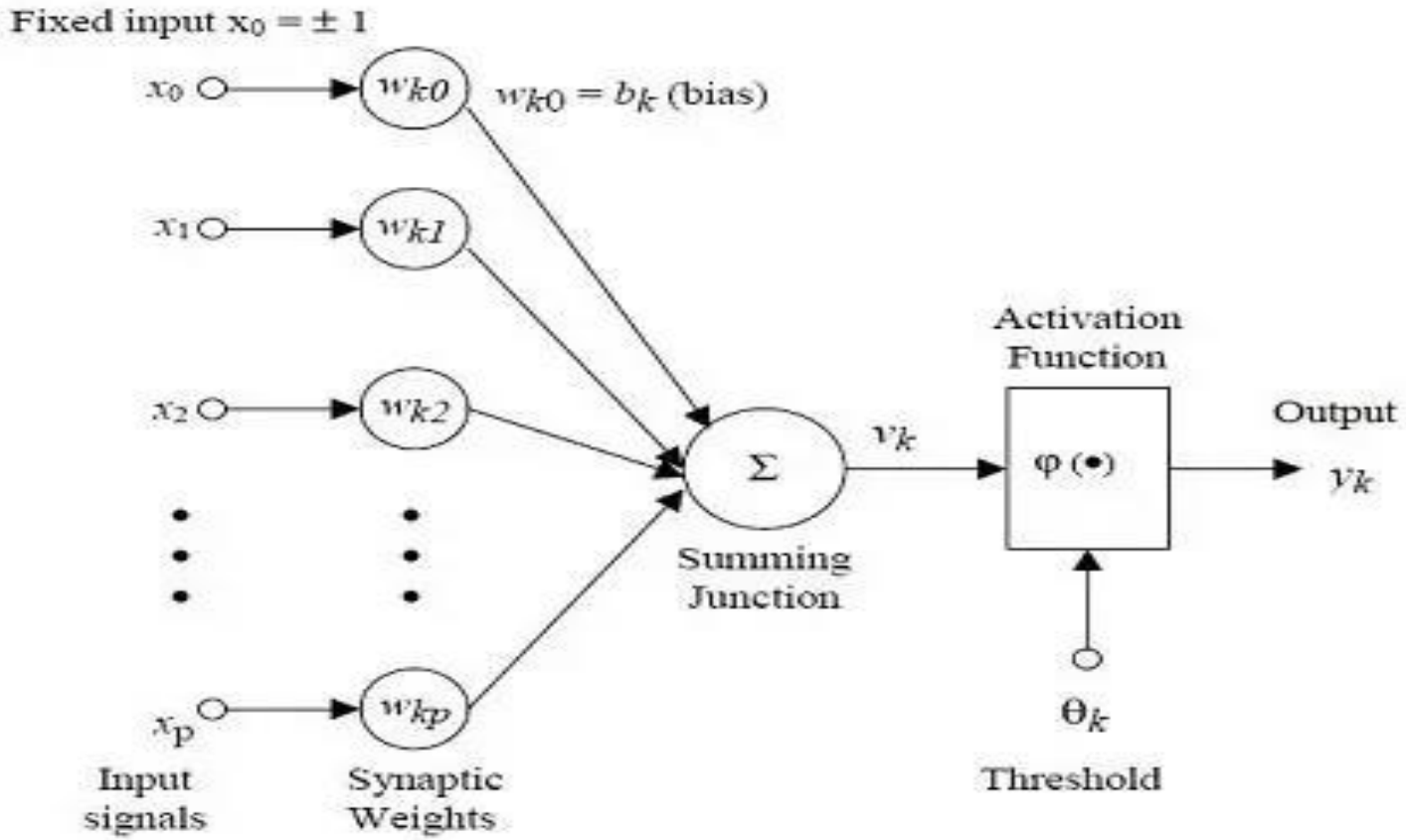
ANN Component

- To build a good Artificial Neural Network (ANN), you will need the following ingredients:

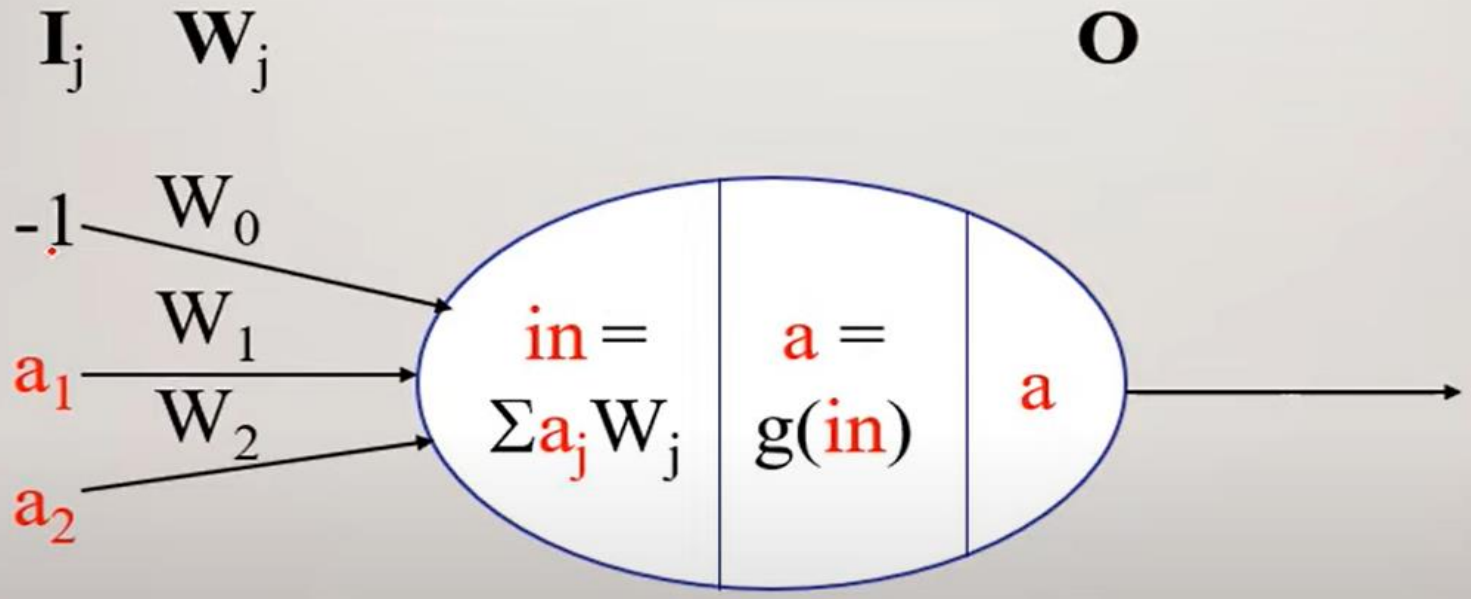
Artificial Neurons (processing node) composed of:

- (many) input neuron(s) connection(s) (dendrites)
- a computation unit (nucleus) composed of:
  - a linear function (ax+b)
  - an activation function (equivalent to the synapse)
  - an output (axon)

ANN Architecture

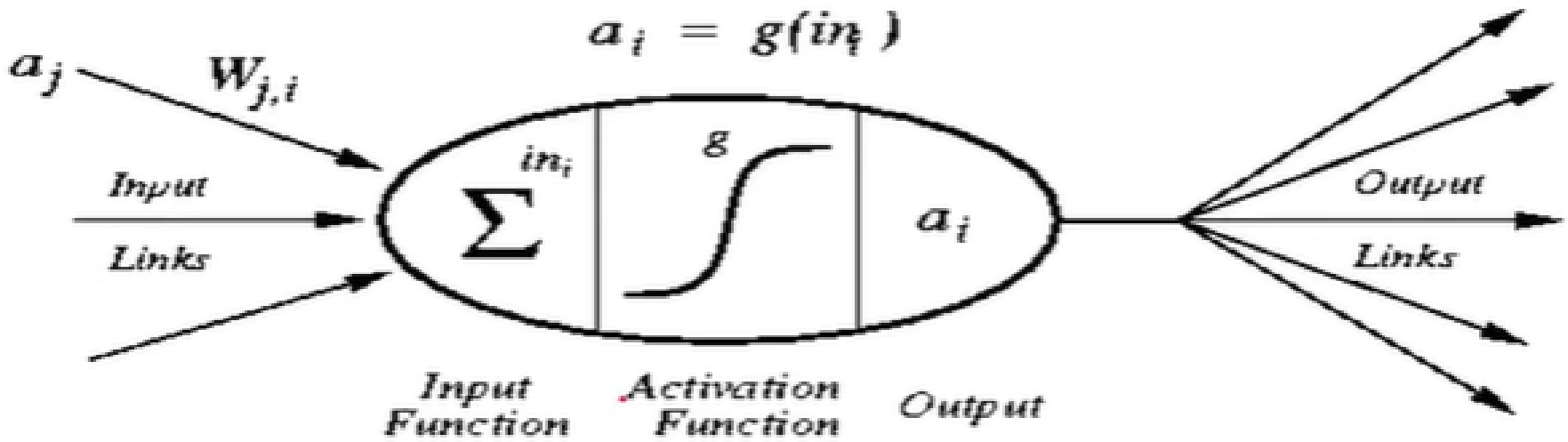


ANN Architecture (cont.,)



$$a = g(-W_0 + W_1 a_1 + W_2 a_2) \qquad g(in) = \begin{cases} 0, & in<0 \\ 1, & in>0 \end{cases}$$

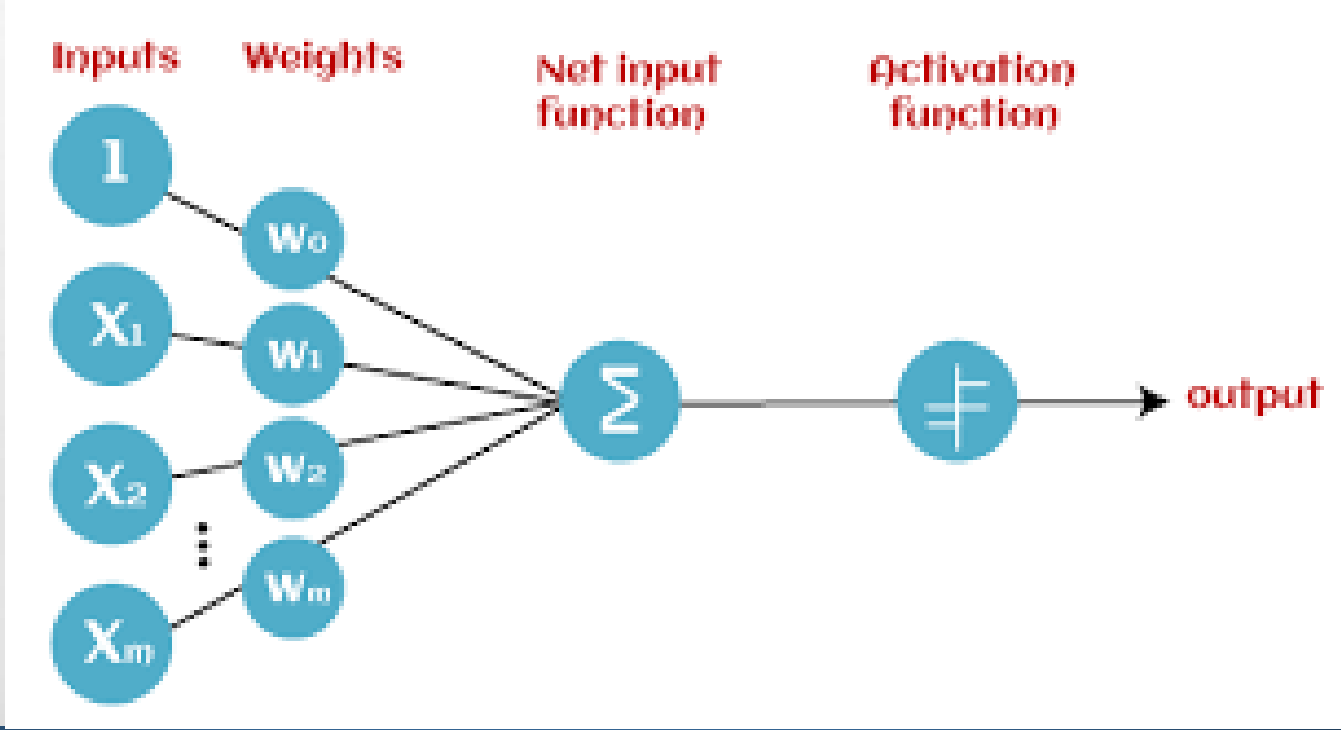
ANN Architecture (cont.,)
Modeling a Neuron



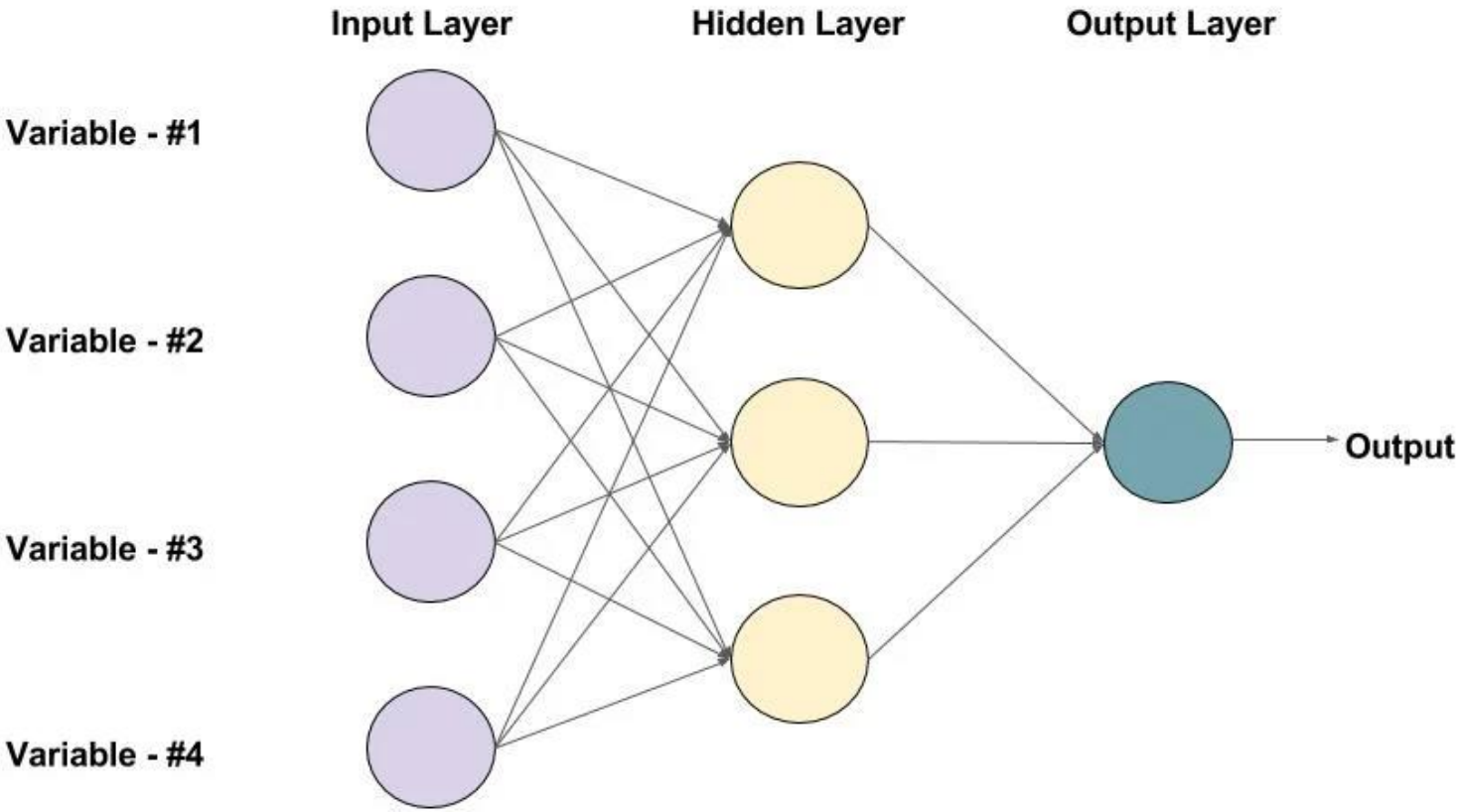
$$in_i = \sum_j W_{j,i} a_j$$

- a_j :Activation value of unit j
- $w_{j,I}$:Weight on the link from unit j to unit i
- in_I :Weighted sum of inputs to unit i
- a_I :Activation value of unit i
- g :Activation function

Neural Network Models
1- Perceptron



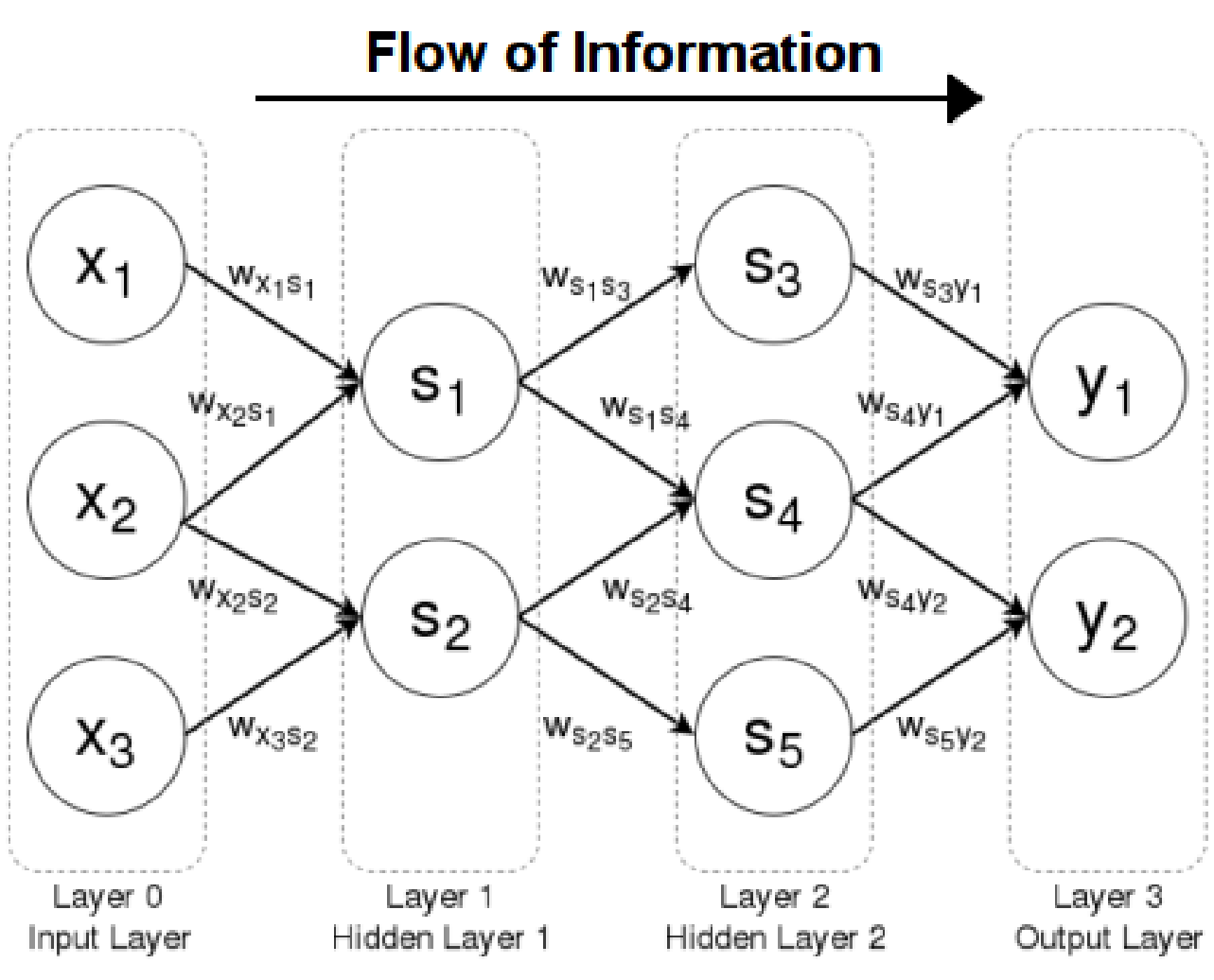
Neural Network Models
2- Feed Forward Neural Network(FFNN)



An example of a Feed-forward Neural Network with one hidden layer (with 3 neurons)

Neural Network Models

2- Feed Forward Neural Network(FFNN)



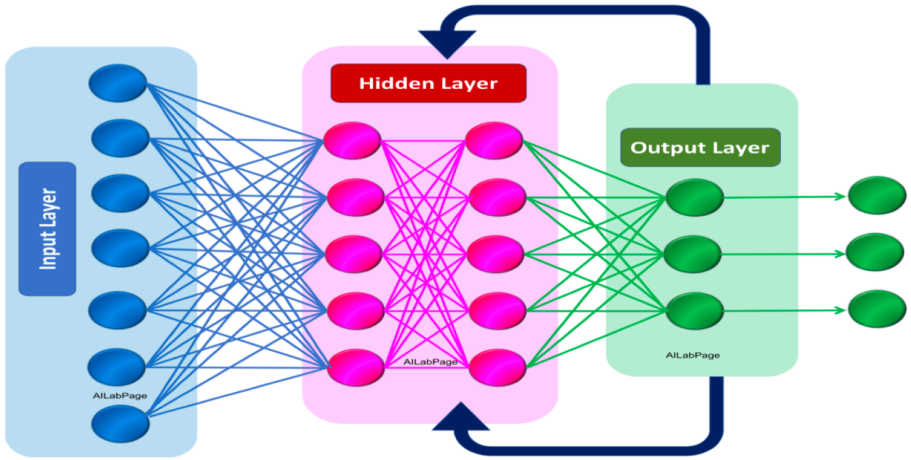
Neural Network Models
2- Feed Forward Neural Network(FFNN)

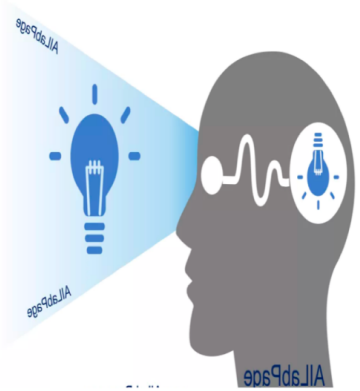
- **Hidden layer** (s)
- **Feedforward** (output links only connected to input links of the next layer).
- **Complex** (non linear function can be represented).

Neural Network Models
3- Recurrent Neural Network(RNN)

Recurrent Neural Networks

Deep Learning – Introduction to Recurrent Neural Networks



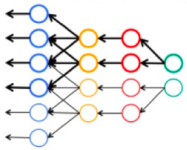


01 Useful
Useful for tasks that are dependent on a sequence of a successive states.

02 Training
The network can be trained by backpropagation.

03 Memory
The network has a form of short-term memory.

04 Simplicity
Simple recurrent network (SRN) has a similar form of short-term memory.



Backpropagation through time

BPTT – An algorithm used for updating weights in the recurrent neural network. to minimize the error of the network outputs. For every recurrent network there is a feedforward network with identical behavior.

Back Activation

A recurrent connection feeds back activation that will affect the output from the network during subsequent iterations.

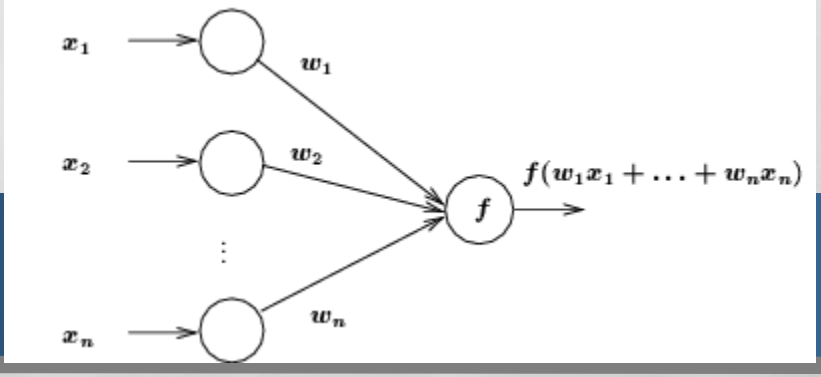


Neural Network Models
3- Recurrent Neural Network(RNN)

- No restriction in connections.
- More difficult to predict or understand.
- Apps: voice to text

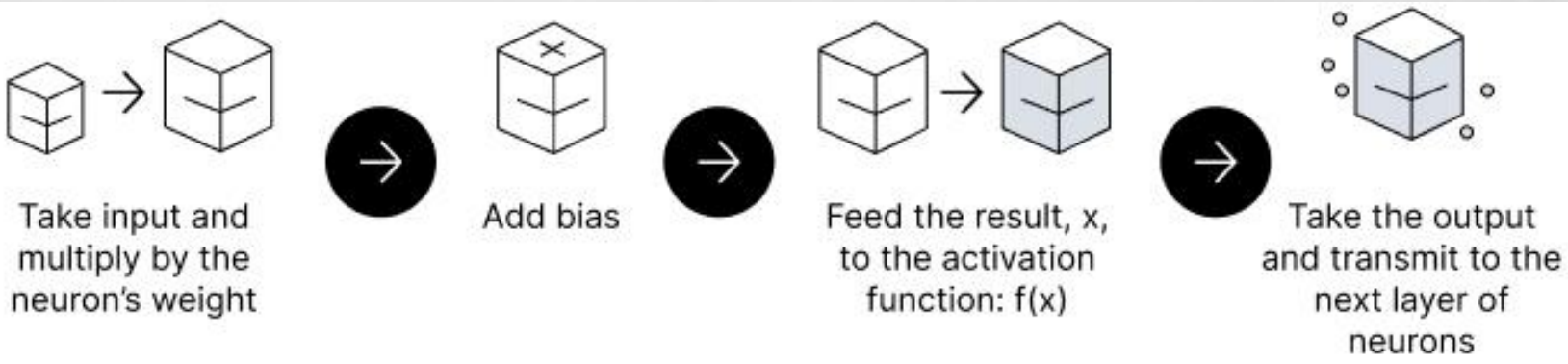
ANN Neuron Models

- Node input
  - Each node has one or more inputs from other nodes, and one output to other nodes
  - Input/output values can be
    - Binary {0, 1}; bipolar {-1, 1}; or continuous (bounded or not)
- Weighted sum of inputs
  - $$\text{output} = f(net) \text{ where } net = \sum_{i=1}^{n} w_i x_i$$



Feedforward vs. Backpropagation

- Those two essential terms describing the movement of information—feedforward and backpropagation.
- **Feedforward Propagation** - the flow of information occurs in the forward direction. The input is used to calculate some intermediate function in the hidden layer, which is then used to calculate an output.

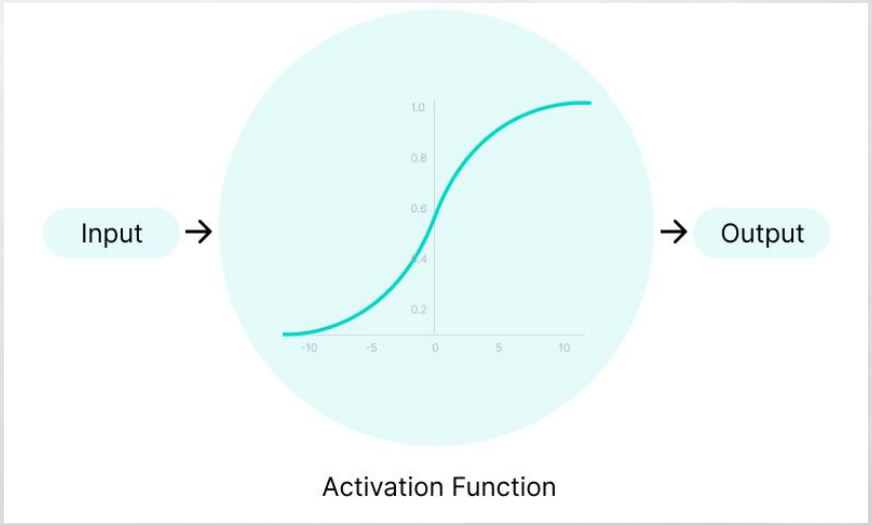


Feedforward vs. Backpropagation

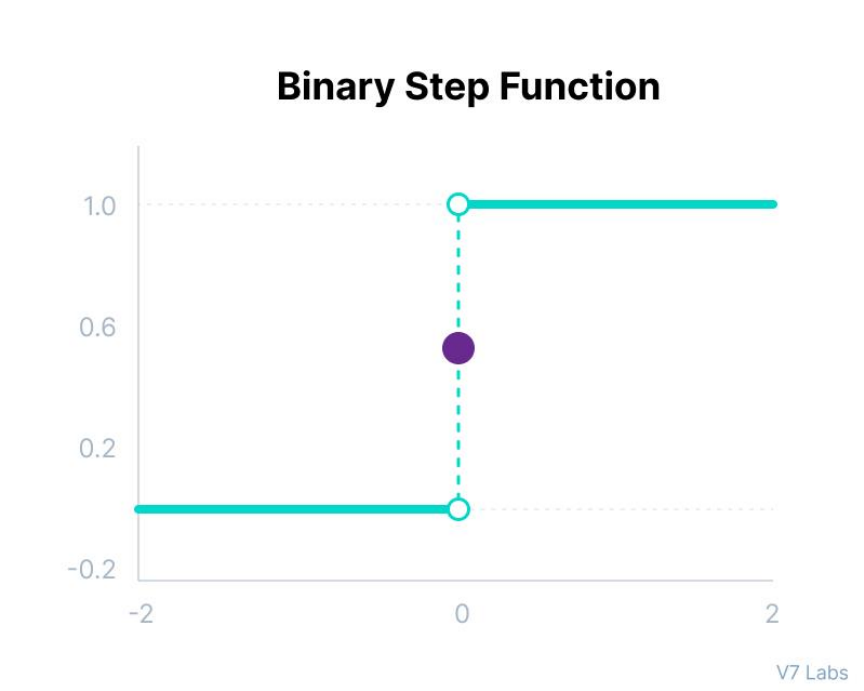
- **Backpropagation** - the weights of the network connections are repeatedly adjusted to minimize the difference between the actual output vector of the net and the desired output vector.

Activation Function

- **Activation Function** helps the neural network to use important information while suppressing irrelevant data points.
- **Why do Neural Networks need it?**
- the purpose of an activation function is to add non-linearity to the neural network.



Activation Function (cont.,)

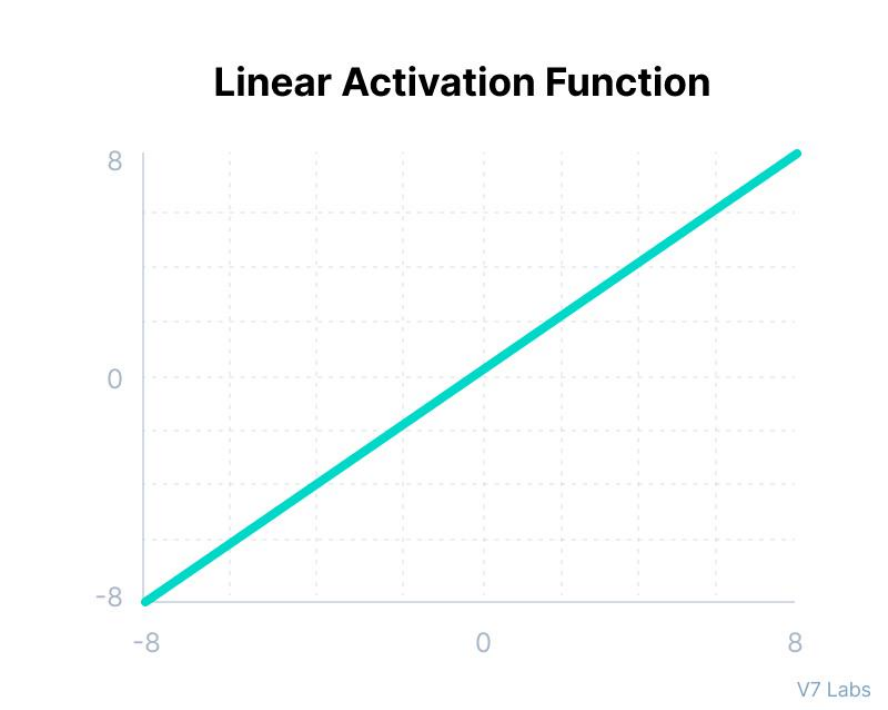


Binary step

$$f(x) = \begin{cases} 0 & for\ x < 0 \\ 1 & for\ x \geqslant 0 \end{cases}$$

Activation Function (cont.,)

- **Linear Activation Function**



Linear

$$f(x) = x$$

Activation Function (cont.,)

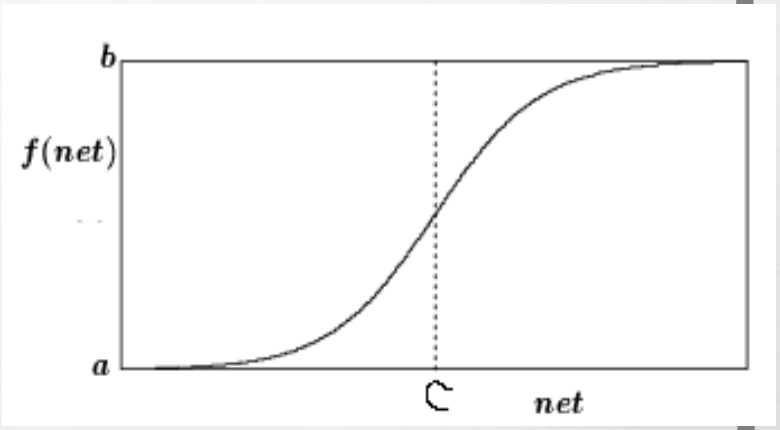


Sigmoid / Logistic

$$f(x) = \frac{1}{1 + e^{-x}}$$

Activation Function (Cont.,)

- **Sigmoid function**
- S-shaped
- Continuous and everywhere differentiable
- Rotationally symmetric
- Asymptotically approaches saturation points
- Examples:
  - Logistic function $$o_j = \frac{1}{1+e^{-net_j}}$$
  - Hyperbolic tangent $$o_j = \frac{e^{net_j} - e^{-net_j}}{e^{net_j} + e^{-net_j}}$$



Neural Network Learning

- Real neural learning
  - Synapses change size and strength with experience.
  - **Hebbian learning**: When two connected neurons are firing at the same time, the strength of the synapse between them increases.

    "Neurons that fire together, wire together."
- ANN learning
  - Construct the NN by training using samples
  - Training is primarily to change the weights
  - We will look at two particular training methods
    - **Perceptron**
    - **backpropagation**

The first NN &Boolean Functions

Table.1 Logical AND Operation

Inputs		Output
A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

Table.2 Logical OR operation

Inputs		Outputs
A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

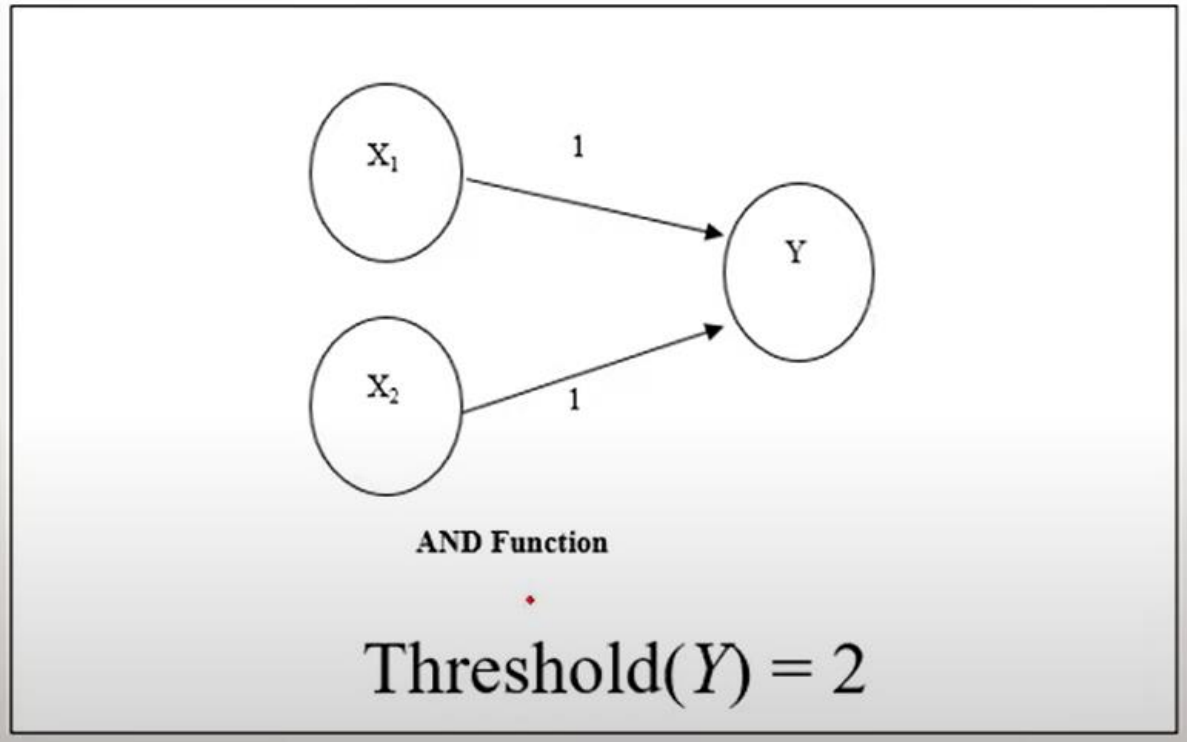
A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

XOR

A	B	Q
0	0	1
0	1	0
1	0	0
1	1	1

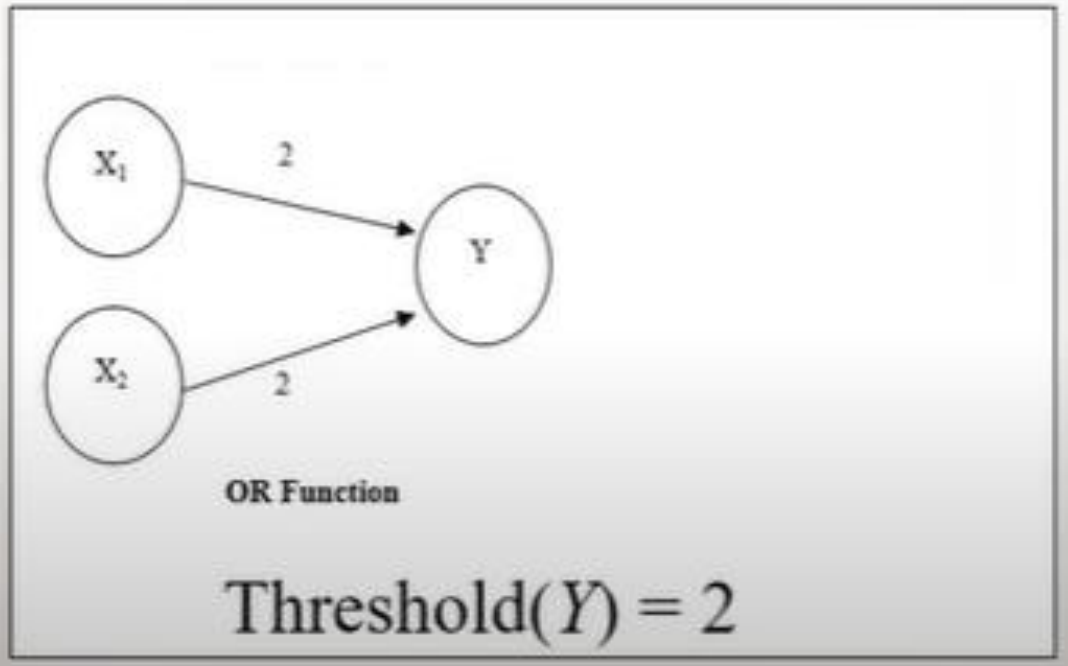
XNOR

The First Neural Network



AND		
X1	X2	Y
1	1	1
1	0	0
0	1	0
0	0	0

The First Neural Network



OR		
X1	X2	Y
1	1	1
1	0	1
0	1	1
0	0	0

Assignment

- Define the different types of activation function, and mention when to use each one according to its pros and cons?

Review Questions

- Explain the components of the artificial neural networks, and describe how it works?
- Differentiate between feedforward neural networks and recurrent neural networks?
- Differentiate between feedforward backpropagation in neural networks?
- What is the main purpose of the activation function?
- Explain the Hebbian learning.
- Explain the perceptron neural network.
- Why perceptron Cannot learn exclusive-or?

Questions ?

Thank You